

The CIA Triad Under Ransomware Attacks: A Study on Database Compromise, Data Integrity, and Cloud-Assisted Detection

**Written to fulfill the 1st task of
Introduction to Cybersecurity**



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PREFACE

Praise be to Allah SWT, the Almighty and Most Merciful, for His blessings and guidance, which have enabled the writer to complete this paper entitled “The CIA Triad Under Ransomware Attacks: A Study on Database Compromise, Data Integrity, and Cloud-Assisted Detection.” This paper is written as a continuation of the writer's previous work on the role of the CIA Triad in Cloud Computing, DBaaS, and Big Data, narrowing the discussion into a single, concrete, and currently developing threat: ransomware attacks against database systems.

During the preparation of this paper, the writer gained deeper insight into how a single class of attack, ransomware, is capable of simultaneously undermining all three pillars of the CIA Triad, confidentiality through data exfiltration, integrity through data deletion or corruption, and availability through service disruption. It is expected that this paper can help readers, especially students in the field of information technology, to better understand how real-world academic research characterizes and defends against this evolving threat.

The writer would like to express sincere gratitude to the lecturers who have provided guidance and knowledge during the learning process, as well as to friends and colleagues who offered support, discussions, and motivation throughout the completion of this paper.

The writer realizes that this paper is still far from perfect. Therefore, constructive suggestions and feedback are highly appreciated for future improvement. Hopefully, this paper can be useful for both the writer and the readers as part of a continuous learning process in the field of information technology.

Depok, August 2026

Author

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CHAPTER I

INTRODUCTION

1.1 Background

Databases sit at the core of virtually every digital service, storing the operational and personal data organizations depend on. As more of this data moves into cloud computing and Database-as-a-Service (DBaaS) platforms, the potential impact of a single security failure grows accordingly: one exposed database server can translate directly into a large-scale breach, a business shutdown, or a costly extortion attempt. Among the many threats facing these environments, ransomware has emerged as one of the most damaging and persistent, evolving from a threat that once targeted individual desktop computers into a sophisticated, multi-billion-dollar criminal industry that now increasingly targets servers, databases, and cloud infrastructure directly.

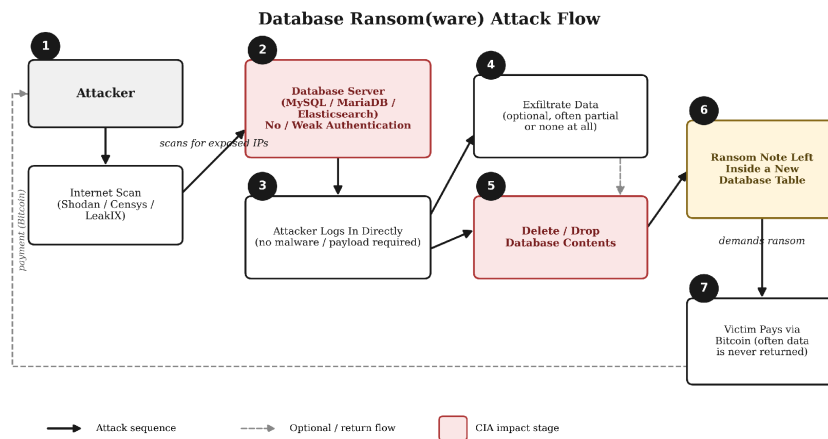


Figure 1.1 Database Ransomware Attack Flow (Source: Adapted by the writer from Van Liebergen et al., NDSS 2025)

Unlike traditional ransomware, which typically infiltrates a victim's device through phishing emails or malicious downloads before encrypting local files, a distinct and increasingly prevalent variant, commonly referred to as database ransom(ware) attacks, targets database servers directly over the network. Attackers scan the internet for database services that lack proper authentication or use weak, default credentials, log in, exfiltrate or delete the stored data, and leave behind a ransom note demanding payment for its return. Recent academic research shows that this class of attack is far from a fading threat: newly infected servers continue to appear by the thousands every month, and the operations behind them are increasingly organized, profit-driven, and in some cases linked to state-sponsored actors.

This paper builds directly on the writer's earlier study of the CIA Triad, Confidentiality, Integrity, and Availability, in the context of Cloud Computing, DBaaS, and Big Data. Rather than discussing the triad in general terms, this paper narrows the focus to a single, concrete threat, ransomware attacks on databases, and examines how such attacks simultaneously compromise all three pillars of the triad: confidentiality through data exfiltration, integrity through data deletion or corruption, and availability through service disruption. By grounding the discussion in recent, peer-reviewed research, this paper aims to provide a more concrete and evidence-based understanding of how ransomware operates against modern database systems, and how emerging, cloud-assisted defenses attempt to counter

1.2 Objective

The purpose of this paper is to analyze how ransomware attacks against database systems compromise the Confidentiality, Integrity, and Availability of data. This paper discusses the anatomy of database ransom attacks, the scale and organization of the threat actors behind them, the broader empirical patterns of ransomware-induced data disruption, and an emerging cloud-assisted detection architecture designed to counter increasingly evasive ransomware variants. Through this discussion, the paper aims to help readers understand not only the theoretical relevance of the CIA Triad but also how it manifests in a real, currently active class of cyberattack.

1.3 Problem Domain

This paper focuses on ransomware attacks that specifically target database servers, supported by a broader look at how ransomware disrupts data integrity in general and how cloud-assisted architectures attempt to detect evasive ransomware variants. The scope covers three interrelated aspects: (1) how attackers identify and compromise poorly secured database servers, (2) how the resulting data disruption undermines the integrity of victim systems at a broader, empirical scale, and (3) how a cloud-assisted detection architecture attempts to identify ransomware activity even when the ransomware employs advanced evasion techniques, while acknowledging that this architecture was not originally designed for database environments. The discussion is grounded in three recent academic studies published between 2024 and 2026, ensuring that the analysis reflects the current state of the threat landscape rather than outdated assumptions.

1.4 Writing Methodology

The writing method for this paper uses short journals and literature studies. The author collects information from peer-reviewed academic publications, including papers presented at the Network and Distributed System Security (NDSS) Symposium, the IEEE/ACM International Conference on Software Engineering (ICSE), and recent preprints on cloud-based ransomware defense, supplemented by reputable cybersecurity industry reports.

1.5 Writing Framework

Systematics that the author uses in this ISAS investigation is arranged as follows:

- **CHAPTER I: INTRODUCTION** - Describes the background, objectives, scope, methodology, and writing framework of the paper.
- **CHAPTER II: BASIC THEORY** - Recaps the CIA Triad and introduces the theoretical foundation of ransomware and database ransom attacks.
- **CHAPTER III: PROBLEM ANALYSIS** - Presents findings from three recent journals covering the characterization of database ransom attacks, the empirical study of data disruption, and cloud-based detection systems, including an honest discussion of scope limitations.
- **CHAPTER IV: CONCLUSION AND SUGGESTION** - Summarizes the findings and offers practical recommendations.

CHAPTER II

BASIC THEORY

2.1 CIA Triad Recap

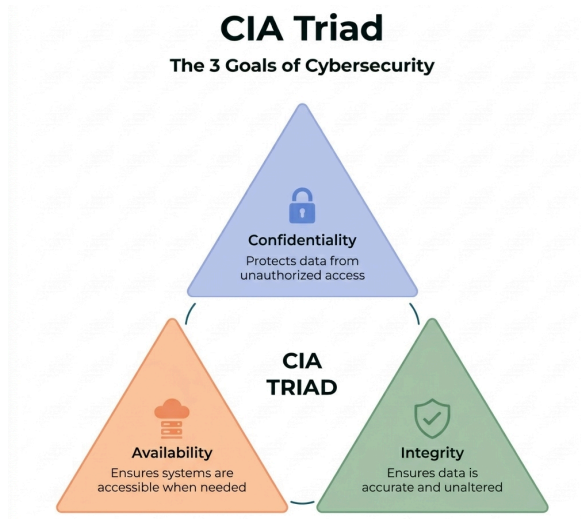


Figure 2.1 Goals of Cybersecurity (Source:futureense)

As established in the writer's previous paper, the CIA Triad, Confidentiality, Integrity, and Availability, remains the foundational model for evaluating information security. In the context of this paper:

- **Confidentiality** refers to restricting data access to authorized entities only. In database ransom attacks, confidentiality is threatened when attackers exfiltrate data before deleting it, or threaten to leak it publicly.
- **Integrity** refers to the trustworthiness and correctness of data throughout its lifecycle. Ransomware directly attacks integrity by deleting, corrupting, or encrypting data without authorization.
- **Availability** refers to ensuring that data and services remain accessible to authorized users. A successful ransomware attack renders databases and dependent applications unavailable until the data is restored or the ransom is resolved.

What makes ransomware a particularly instructive case study for the CIA Triad is that a single attack simultaneously threatens all three pillars, unlike many other attack types that primarily target only one.

2.2 Ransomware: Definition and Evolution

Ransomware is a class of malicious activity in which an attacker denies a victim access to their own data or systems and demands payment, typically in cryptocurrency, in exchange for restoring that access. Historically, ransomware encrypted files on a victim's device and requested a decryption key in return for payment. Over time, ransomware groups adopted “double extortion” tactics, exfiltrating data before encrypting it so that victims face both a loss of access and a threat of public data leakage.

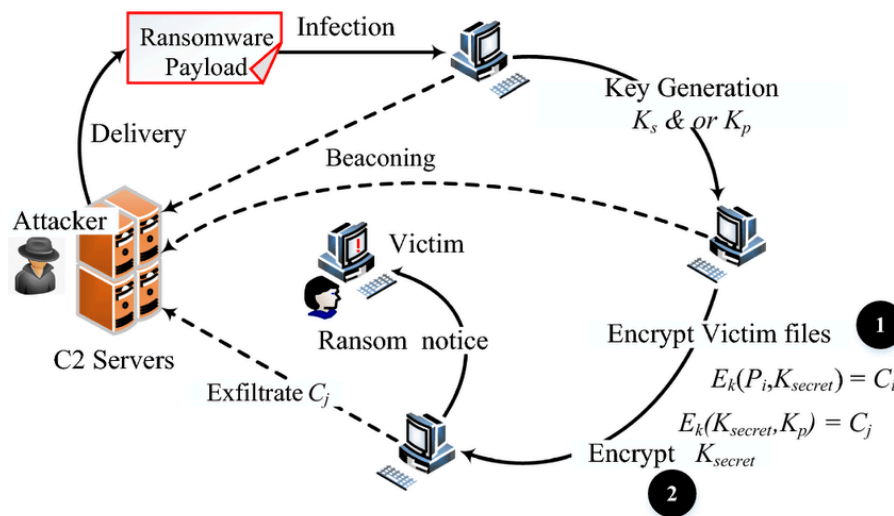


Figure 2.2 Typical Ransomware Attack Process (Source: ResearchGate)

More recently, ransomware campaigns have increasingly combined privilege escalation with sophisticated evasion strategies, such as intermittent encryption, low-entropy encryption, and imitation attacks, that allow malware to bypass traditional detection systems relying on I/O-pattern or statistical analysis. These evolved variants are collectively referred to in recent literature as Privilege-Escalated Evasive Ransomware (PEER)

2.2.1 Privilege-Escalated Evasive Ransomware (PEER)

Privilege-Escalated Evasive Ransomware (PEER) executes through a streamlined four-stage mechanism:

- **Infiltration & Privilege Escalation:** Upon gaining initial access, PEER exploits local vulnerabilities or misconfigurations to achieve elevated privileges (e.g., SYSTEM or kernel level), enabling it to neutralize security software and gain unrestricted file access.

- **Stealth & Evasion Execution:** The malware bypasses behavioral detection using evasion techniques, such as *intermittent encryption* (encrypting partial file blocks), *low-entropy schemes* (mimicking benign data entropy), and *process imitation* (mimicking legitimate system behavior).
- **Data Exfiltration:** Utilizing its high privileges, the malware exfiltrates sensitive files to external command-and-control (C2) servers to facilitate double extortion.
- **Targeted Impact & Extortion:** PEER rapidly encrypts key assets, terminates shadow copies to obstruct local recovery, and displays the ransom demand.

2.3 Database Ransomware Attacks

A distinct category of ransomware, known as database ransom(ware) attacks or malwareless ransomware, targets database servers directly rather than end-user devices. Instead of relying on malware or social engineering, attackers scan the internet for database servers (commonly MySQL, MariaDB, and Elasticsearch) that lack authentication or use weak, default credentials. Once inside, the attacker deletes the database contents and leaves a ransom note, often in the form of a newly created table, demanding payment for the data's return.

This approach differs meaningfully from traditional ransomware in several ways: it requires no malware binary, it exploits misconfiguration rather than software vulnerabilities, and it is fully automatable at internet scale. These characteristics make database ransom attacks especially relevant to the discussion of Cloud Computing and DBaaS security, since misconfigured, unauthenticated access is a well-documented risk in shared responsibility cloud models.

2.4 Cloud-Assisted Ransomware Detection

As ransomware becomes more evasive, defensive research has also evolved. Traditional detection approaches relying on I/O-pattern monitoring or statistical entropy analysis at the operating system level are increasingly unreliable, particularly against privilege-escalated attackers who can disable or tamper with local monitoring agents once they gain control of a device. A newer defensive paradigm addresses this problem by offloading both detection and recovery to the cloud: rather than trusting security software running on a potentially compromised endpoint, data snapshots are periodically replicated to an isolated cloud environment where analysis takes place beyond the attacker's reach.

It is important to note that this cloud-assisted approach, discussed in detail in Chapter III, has so far been developed and evaluated primarily for endpoint devices and virtual desktop infrastructure (VDI), protecting common file formats such as documents, spreadsheets, and archives, rather than for database servers or Database-as-a-Service (DBaaS) platforms directly. Nevertheless, the underlying architectural principle, isolating detection logic in a cloud environment that remains trustworthy even when the monitored system is compromised, is conceptually relevant to broader discussions of cloud and database security, and is examined critically in that light in Chapter III.

CHAPTER III

PROBLEM ANALYSIS

This chapter analyzes three recent academic studies that, together, illustrate how ransomware attacks compromise the Confidentiality, Integrity, and Availability of database systems, and how modern research is responding to this threat.

3.1 Characterizing Database Ransom Attacks

This paper presents the first systematic academic study of database ransom(ware) attacks. The authors analyzed 23,736 ransom notes collected from 60,427 compromised Elasticsearch, MySQL, and MariaDB servers over a three-year period (May 2021–April 2024), and additionally deployed their own honeypot databases to observe live attacks first-hand.

Source: Van Liebergen, K., Gomez, G., Matic, S., & Caballero, J. (2025). *All your (data)base are belong to us: Characterizing Database Ransom(ware) Attacks*. Network and Distributed System Security (NDSS) Symposium 2025, San Diego, CA, USA.

3.1.1 Confidentiality and Availability Impact

The study's honeypot experiments revealed that unauthenticated database servers are infected remarkably quickly, on average within 14 hours of being connected to the internet, and are frequently reinfected afterward, indicating that these attacks are largely automated. The researchers also found that weak authentication is over one hundred times more common on Elasticsearch servers than on MySQL servers, primarily because Elasticsearch only enabled strong default credentials starting from version 8.0 (released in February 2022), and adoption of this newer version remains slow. This finding directly illustrates a confidentiality failure rooted in default configuration and slow software adoption, a pattern highly relevant to the shared responsibility model discussed in the writer's earlier CIA Triad paper.

Database	Events	IP	CC	ASN	Notes	Addr.
ElasticSearch	254,629	40,663	123	1,926	17,391	86
MySQL	43,678	18,224	106	1,198	5,790	316
MariaDB	3,939	1,906	75	491	661	191
All	302,246	60,427	139	2,646	23,736	401

Table II: LeakIX dataset summary.

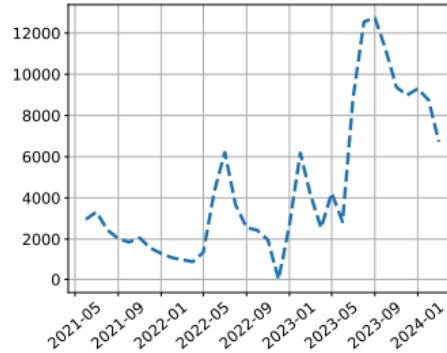


Figure 3.1 Monthly Number of Infected Database IPs, 2021–2024 (Source: Van Liebergen et al., NDSS 2025)

As shown in Figure 3.1, the number of monthly infected IP addresses grew substantially from May 2021 through mid-2023, and although it declined somewhat afterward, infections remained significantly higher in early 2024 than they were before August 2023, over 6,000 newly infected servers were recorded in March 2024 alone, a 60% increase compared to one year earlier. This trend demonstrates that database ransom attacks are not a fading or historical threat but an ongoing and, in some respects, worsening problem.

3.1.2 Who Is Behind the Attacks?

To determine who was responsible for these attacks in the absence of any malware binary to analyze, the researchers designed a novel clustering approach that groups ransom notes into campaigns based on textual similarity, then merges related campaigns into threat groups using shared indicators of compromise (such as Bitcoin addresses) and Bitcoin blockchain transaction analysis

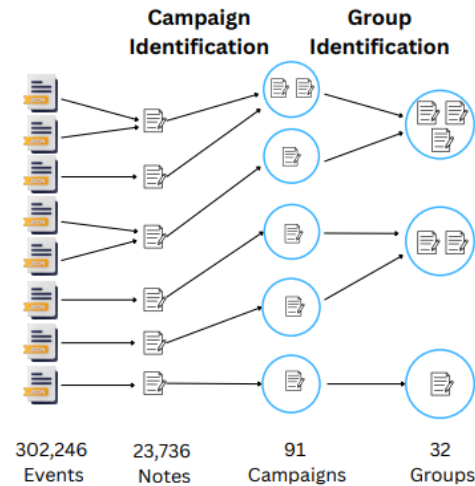


Figure 3.2 Overview of the Database Ransom Attack Clustering Approach (Source: Van Liebergen et al., NDSS 2025)

As illustrated in Figure 3.2, this approach reduced 302,246 raw infection events down to 23,736 unique ransom notes, then grouped them into 91 campaigns run by just 32 distinct threat groups. Strikingly, a single dominant group was found to be responsible for 76% of all infected servers and over 90% of the estimated financial revenue. This concentration of criminal activity in the hands of a small number of highly organized groups underscores that database ransom attacks, while appearing opportunistic and automated at the individual server level, are in fact orchestrated at scale by a small number of sophisticated operators.

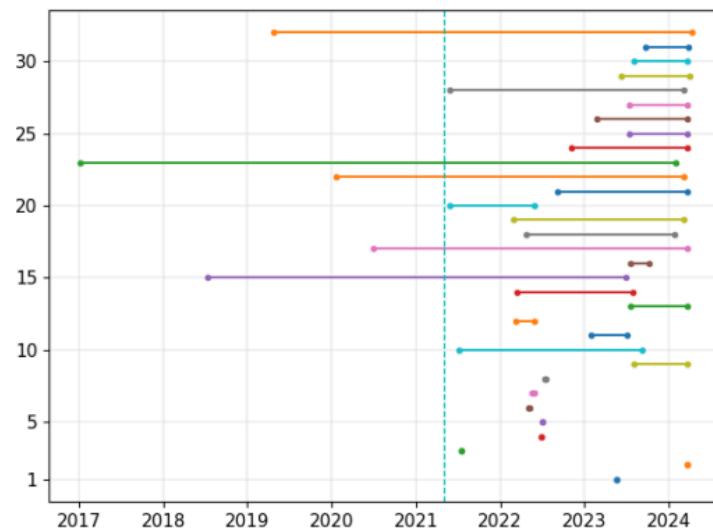


Figure 3.3 Group Lifespan of Database Ransom Threat Groups (Source: Van Liebergen et al., NDSS 2025)

Figure 3.3 further shows that several groups, including the dominant one, have been active since before the study's dataset even began (as early as 2017 for one group), demonstrating that this is a long-running criminal enterprise rather than a short-lived trend. Eighteen of the thirty-two groups identified were still recording new infections in 2024, confirming that the threat remains active today.

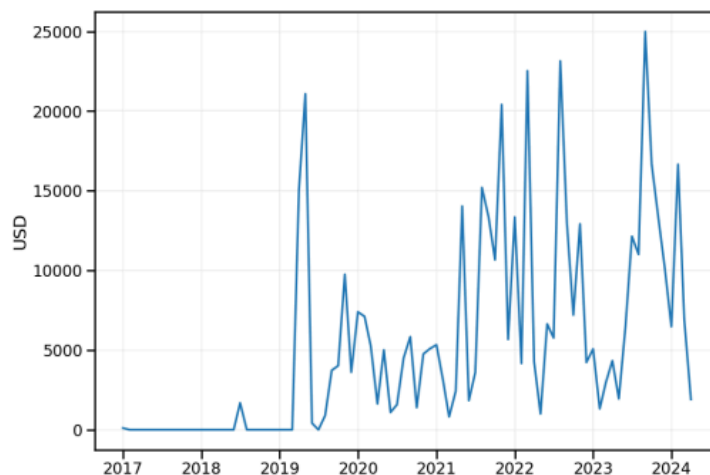


Figure 3.4 Total Monthly Revenue Across All Groups (Source: Van Liebergen et al., NDSS 2025)

Interestingly, as shown in Figure 3.4, the total measured revenue across all groups, approximately \$498,000 over the study period, is relatively modest compared to other forms of cybercrime. The researchers attribute this to deliberately small ransom amounts (94.4% of payments were under \$1,000), likely intended to make payment more attractive to uncertain victims. However, the researchers caution that this figure is very likely a substantial underestimate, since internet-scanning engines are known to detect only a small fraction of all infected servers in similar attacks.

3.1.3 Integrity: Can Victims Actually Recover Their Data?

Perhaps the most sobering finding of this study concerns data integrity and recoverability. Because database ransom attacks delete data rather than encrypt it, recovery is only possible if the attacker actually exfiltrated a copy before deletion, and is willing to return it after payment. The researchers found that their honeypots' data was only ever partially exfiltrated (or not exfiltrated at all in some cases), and that multiple publicly reported cases showed victims who paid the ransom but never received their data back. This strongly suggests that many database ransom attacks function as scams with no genuine intention of returning stolen data, a critical insight for any organization considering whether to pay a ransom.

3.1.4 Link to Nation-State Activity

The study also uncovered a financial link between the dominant threat group and a Bitcoin address previously attributed by the U.S. Department of Defense to North Korea (DPRK), as well as a link to a separate 2019 campaign that had targeted Git repositories on GitHub, GitLab, and BitBucket using leaked credentials. While the researchers are careful not to draw an overly conclusive attribution, this finding illustrates that even seemingly small-scale, opportunistic attacks on misconfigured databases may ultimately be connected to much larger, state-linked cybercriminal operations.

3.2 Data Disruption: The Integrity Dimension

While the NDSS study above focuses specifically on database servers, this second study broadens the analysis to ransomware's impact on data integrity across a much larger and more diverse dataset. The researchers built a real-world dataset named MarauderMap, consisting of 7,796 active and unique ransomware samples drawn from 95 distinct malware families, and executed every sample inside isolated testbeds to capture six categories of runtime behavior, including API calls, file I/O access, and network traffic, resulting in nearly 1.98 terabytes of behavioral log data.

The scale and diversity of this dataset make it particularly valuable for understanding data disruption as a distinct, integrity-focused dimension of ransomware behavior, separate from the confidentiality-oriented exfiltration and availability-oriented downtime discussed in the previous section. It is important to note that this study is not specific to databases; it examines ransomware behavior in general across a broad range of file systems and environments. Its relevance to this paper is therefore conceptual rather than direct: it supplies large-scale empirical evidence for the integrity dimension of the CIA Triad, complementing the narrower, database-specific findings in Section 3.1 rather than extending them directly.

The practical value of this research lies in its scale: because prior ransomware studies were often limited to a handful of samples or a single ransomware family, findings could not easily be generalized. By analyzing thousands of active, real-world samples across dozens of families, this study provides a much stronger empirical foundation for designing detection and mitigation strategies that generalize across the broader ransomware ecosystem, rather than being narrowly tuned to a single, soon-to-be-outdated variant.

Source: Hou, Y., Guo, L., Zhou, C., Xu, Y., Yin, Z., Li, S., Sun, C., & Jiang, Y. (2024). *An Empirical Study of Data Disruption by Ransomware Attacks*. Proceedings of the IEEE/ACM 46th International Conference on Software Engineering (ICSE '24), Lisbon, Portugal.

3.3 Cloud-Based Detection and Its Limitations

The two studies discussed above characterize the scale and impact of the threat; this final study proposes a concrete, cloud-based defense against one of the most difficult challenges in modern ransomware detection: Privilege-Escalated Evasive Ransomware (PEER). PEER variants combine elevated system privileges with sophisticated evasion strategies, such as intermittent encryption, low-entropy encryption, and imitation attacks, that allow them to defeat traditional detection systems relying on I/O-pattern monitoring or statistical entropy analysis, since a privileged attacker can simply disable or tamper with local monitoring agents.

Source: Kim, B. H., Hong, S. M., & Mannan, M. (2026). *Rhea: Detecting Privilege-Escalated Evasive Ransomware Attacks Using Format-Aware Validation in the Cloud*. Preprint, under review.

3.3.1 The Rhea Architecture

Rhea is a cloud-offloaded ransomware defense system that shifts both detection and recovery away from the potentially compromised end device and into an isolated cloud environment. Rather than trusting telemetry gathered on a device that may already be under attacker control, Rhea periodically replicates block-level data snapshots, called mutation snapshots, from edge devices (such as desktops, laptops, or virtual desktop infrastructure) to a cloud backend, where all analysis takes place beyond the attacker's reach.

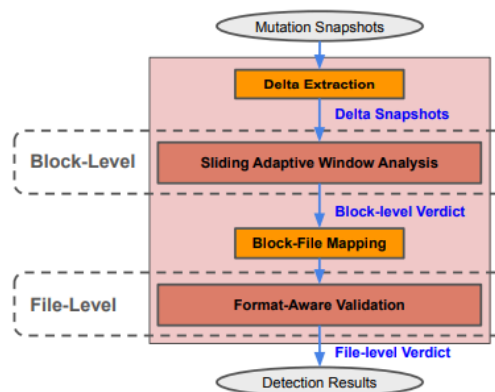


Figure 3.5 Rhea Format-Aware Content-Only Detection Pipeline (Source: Kim, Hong & Mannan, 2026)

As shown in Figure 3.5, the Rhea agent processes these snapshots through a four-stage, dual-level detection pipeline. The first two stages, Delta Extraction and Sliding Adaptive Window Analysis (SAWA), operate at the block level and act as an efficient, high-recall pre-filter that narrows down candidate regions of suspicious data using statistical (chi-squared) analysis, illustrated in Figure 3.6 and Figure 3.7 respectively.

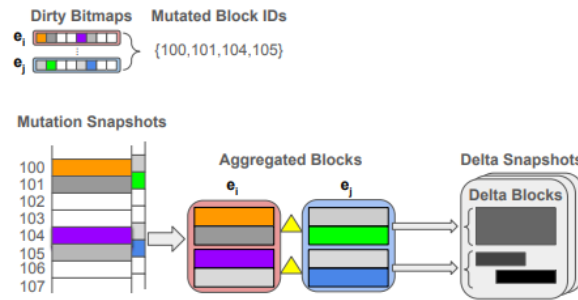


Figure 3.6 Delta Extraction Process in Rhea (Source: Kim, Hong & Mannan, 2026)

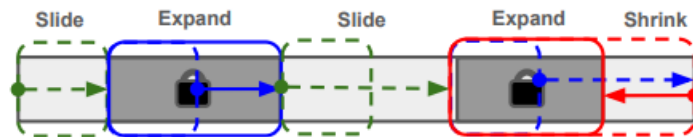


Figure 3.7 Illustration of the Sliding Adaptive Window Analysis (SAWA) Process (Source: Kim, Hong & Mannan, 2026)

The final two stages, Block-File Mapping and Format-Aware Validation (FAV), operate at the file level and make the authoritative detection decision. The core innovation of Rhea lies in FAV: instead of relying on statistical measures of randomness (such as Shannon entropy or chi-squared tests), which become unreliable when ransomware encrypts only small, scattered spans of a file, FAV checks whether a file's content still conforms to the syntactic and semantic structure defined by its own file format specification (for example, whether a .docx file's internal XML and ZIP container structure remains valid). Because even a small amount of encryption typically breaks these structural rules, FAV can reliably detect fine-grained partial encryption that would otherwise evade traditional statistical detectors.

3.3.2 Detection Effectiveness

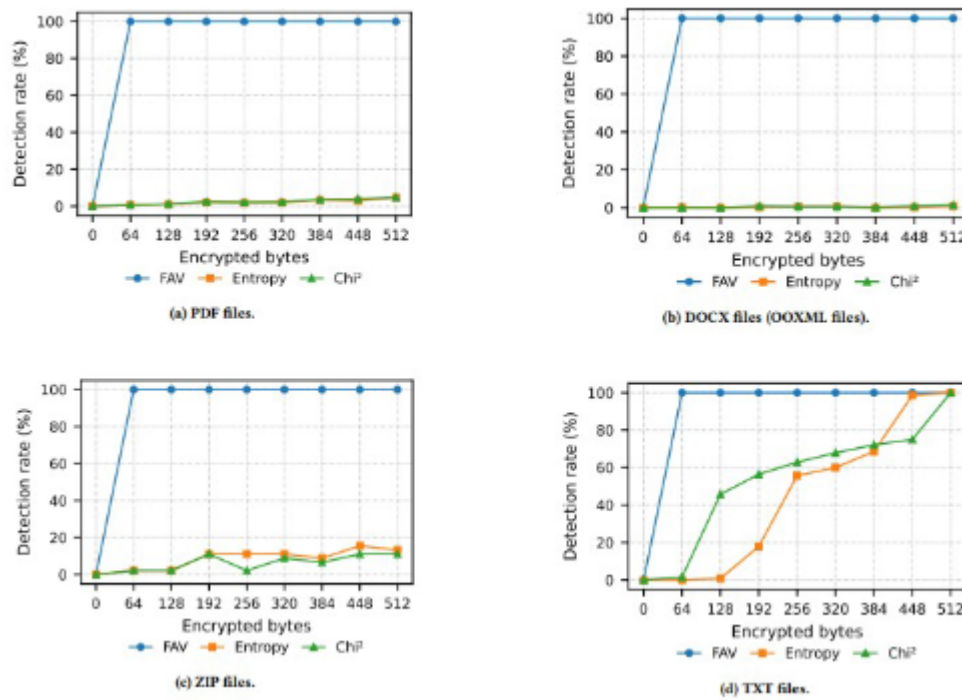


Figure 3.8 Detection Rate Comparison: FAV vs. Entropy vs. Chi-Squared (PDF, DOCX, ZIP, TXT)

(Source: Kim, Hong & Mannan, 2026)

Figure 3.8 compares the detection rate of FAV against traditional entropy-based and chi-squared-based detectors as the amount of encrypted data per write is reduced from 512 bytes down to 64 bytes, across four common file formats. Across all four formats, FAV achieves a 100% detection rate almost immediately once any encryption is present, while the entropy and chi-squared baselines struggle badly, particularly for ZIP files, where they never exceed roughly 15% detection even at 512 encrypted bytes, and for PDF and DOCX files, where they remain close to 0% throughout the entire tested range. This stark contrast demonstrates the practical value of format-aware validation over purely statistical approaches, at least for the office-document and archive formats it was designed to protect.

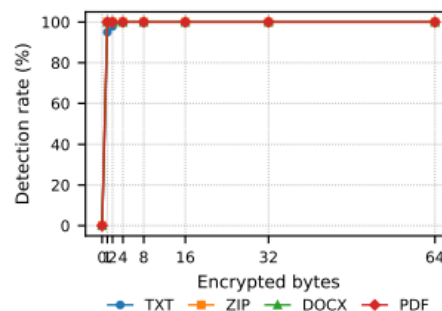


Figure 3.9 Detection Rate Under Extremely Fine-Grained Partial Encryption (Below 64 Bytes) (Source: Kim, Hong & Mannan, 2026)

The researchers pushed this evaluation even further, testing encryption granularities below 64 bytes, down to modifying just a single byte per write. As shown in Figure 3.9, FAV maintains perfect or near-perfect detection across PDF, DOCX, and ZIP formats even at this extreme, with only plain text (TXT) files showing a slight dip to 95% detection at the single-byte level. In a separate real-world evaluation using 78 actual ransomware samples spanning 35 known families (including Babuk, Akira, Medusa, and LockFile), Rhea achieved 100% accuracy, precision, and recall, with zero false positives and zero false negatives.

3.3.3 Relevance and Limitations: Rhea's Position Outside the Database Context

It is important to clarify the actual scope of Rhea before drawing conclusions about its relevance to this paper's discussion of database ransom attacks. Rhea is not a database-specific detection system, nor does it operate within a DBaaS environment. Its target files are common desktop and office document formats, TXT, PDF, ZIP, and OOXML formats such as DOCX, XLSX, and PPTX, along with media files such as JPG, PNG, MP3, and MP4. Its deployment environment is edge devices, including desktops, laptops, tablets, and virtual desktop infrastructure (VDI) commonly used in enterprise settings, not internet-facing database servers of the kind studied in Section 3.1.

The word “cloud” in Rhea's title refers specifically to its detection architecture rather than its target environment. Rather than protecting a database hosted in the cloud, Rhea offloads the detection and recovery process itself to the cloud: block-level data snapshots from an edge device are periodically replicated to a cloud backend, where analysis takes place in an environment the attacker cannot directly tamper with, even after achieving elevated privileges on the original device. This distinction matters because the database ransom attacks discussed in Section 3.1 exploit a fundamentally different attack surface, unauthenticated or weakly authenticated database servers exposed directly to the internet, a scenario Rhea's file-format-based detection was not designed to address.

Given this distinction, Rhea should not be interpreted as a direct technical countermeasure against the database ransom attacks characterized in Section 3.1. Its relevance to this paper lies instead at the architectural level: the underlying principle of offloading security analysis to an isolated, cloud-based environment so that detection cannot be disabled by a compromised endpoint is a design pattern that could, in principle, inform how database and DBaaS providers build monitoring and anomaly-detection layers that remain trustworthy even if a client-side or application-level component is compromised. However, translating

this principle into an actual database-ransom detection mechanism, for example, one that inspects database query patterns or authentication anomalies rather than file-format structure, would require substantial adaptation beyond what Rhea currently provides.

This limitation also reinforces a broader point relevant to the CIA Triad discussion: no single piece of research neatly covers all three pillars for a single, unified system. The three journals discussed in this chapter are complementary but address different layers of the problem, database-level confidentiality and availability failures (Section 3.1), broad empirical patterns of data integrity disruption (Section 3.2), and endpoint-level detection architecture (Section 3.3), rather than forming a single end-to-end pipeline. Acknowledging this gap is itself an important finding: it suggests that comprehensive protection against ransomware in database and cloud environments would require combining database-specific access controls with cloud-offloaded, tamper-resistant detection principles similar to Rhea's, an integration that, to the best of the writer's knowledge, does not yet exist as a unified solution in current academic literature.

CHAPTER IV

CONCLUSION AND SUGGESTION

4.1 Conclusion

This paper has examined database ransom(ware) attacks as a concrete, currently active case study of how a single class of cyberattack can simultaneously compromise all three pillars of the CIA Triad. Drawing on three recent academic studies, the characterization of database ransom attacks by Van Liebergen et al. (NDSS 2025), the large-scale empirical study of ransomware-induced data disruption by Hou et al. (ICSE 2024), and the cloud-offloaded detection system Rhea by Kim et al. (2026), several key conclusions can be drawn.

First, database ransom attacks demonstrate a clear confidentiality failure rooted not in sophisticated exploitation, but in basic misconfiguration and weak or absent authentication, a risk directly tied to the shared responsibility model of cloud computing discussed in the writer's earlier paper. Second, these attacks represent a severe and often irreversible integrity failure: because attackers frequently delete data without a genuine intention or ability to return it, victims who pay the ransom cannot assume their data will be recovered. Third, the scale and organization behind these attacks, dominated by a small number of highly active groups with links to broader cybercriminal and even state-sponsored activity, show that what appears to be an opportunistic, automated attack at the individual server level is, in aggregate, a coordinated and lucrative criminal enterprise.

At the same time, defensive research is evolving in response, though not always in perfect alignment with the database-specific threat. The Rhea system illustrates that cloud-offloaded, format-aware detection can reliably identify even highly evasive, privilege-escalated ransomware that defeats traditional statistical detection methods, but as discussed in Section 3.3.3, its current design targets endpoint files rather than database servers. This gap between database-specific attacks and endpoint-focused defenses shows that no single piece of current research fully addresses confidentiality, integrity, and availability for database and cloud systems at once. Taken together, these findings reinforce that the CIA Triad remains not merely a theoretical framework but a practical lens through which real, currently unfolding cybersecurity threats, and the current limits of the research responding to them, can be understood.

4.2 Suggestion

Based on the analysis presented in this paper, several practical recommendations can be made for organizations operating databases in cloud or DBaaS environments:

- **Eliminate default and weak credentials immediately.** Since the majority of database ransom attacks succeed specifically because of missing or default authentication, organizations should enforce mandatory strong-password policies and disable public network exposure of database ports by default.
- **Keep database software up to date.** The disproportionate targeting of older Elasticsearch versions shows that delayed software updates directly translate into a measurably higher risk of compromise; organizations should adopt automated patch management wherever feasible.
- **Maintain offline, immutable backups and verify recoverability regularly.** Given documented cases where victims paid a ransom but never recovered their data, backups, not ransom payments, should be treated as the primary recovery strategy, and backup integrity should be tested periodically rather than assumed.
- **Treat cloud-assisted, endpoint-focused detection as a complementary layer, not a database solution.** As illustrated by Rhea, cloud-offloaded, format-aware detection offers meaningful resilience for endpoint files, but organizations should not assume it protects database servers; database-specific monitoring, such as query-pattern and authentication-anomaly detection, still needs to be built and adopted separately.
- **Treat confidentiality, integrity, and availability as inseparable in incident response planning.** Because ransomware attacks these three pillars simultaneously, incident response plans should not treat data recovery (availability/integrity) and breach disclosure (confidentiality) as separate workstreams, but as a coordinated response.

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